

- Q.29 Describe the function and working of redundant system valve.
- Q.30 Describe the function and design of ejection seats.
- Q.31 Explain various flame proofing methods.
- Q.32 Give the various features incorporated in parachutes.
- Q.33 Give the function, types and examples of cleaning agents.
- Q.34 What material hose pipes are made on in air crafts and where used in hydraulic systems?
- Q.35 Write short note on working of electro hydraulic servo system.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Define actuators, classify them and explain working of any one of them.
- Q.37 Explain the need and design of Air conditioning and pressurizing systems of aircraft.
- Q.38 Describe briefly the types of landing gears and their design principles.

(20)

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5th Sem / AME Sub : Aircraft Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which factor contributes to the pressure drop in hydraulic systems?
- a) Fluid velocity b) Viscosity
c) Hose pipe roughness d) All of the above
- Q.2 Which of the following is not a component of pneumatic de-icing system?
- a) Air separation system
b) Rubber boots
c) Air distribution system
d) Air-Oil separator
- Q.3 Oxygen tank stores the oxygen in liquid form at extremely
- a) Low temperature b) High temperature
c) Low pressure d) High pressure
- Q.4 Landing gear systems utilize shock absorbers to manage _____ forces.
- a) Compressive b) Tensile
c) Impact d) All of the above

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- Q.5 What is the color code of regular unleaded fuel of aircraft
- a) Red b) Yellow
c) Green d) Black
- Q.6 Which one is not the type of fire protection systems.
- a) Sprinklers b) Fire wall
c) Blankets d) Rated doors
- Q.7 Zero-zero ejection seats are designed to work at
- a) Zero pressure b) Zero altitude
c) Zero temperature d) All of the above
- Q.8 What type of actuator or jack is used to move the landing gear during Retraction?
- a) A spring loaded release
b) An electric motor
c) A hydraulic jack or actuator
d) An oleo pneumatic strut
- Q.9 Cold air units are mainly used in
- a) Aircraft air conditioning
b) Automobile engines
c) Power plants
d) Both 'a' and 'c'
- Q.10 Capsules in pneumatic systems are mainly used for
- a) Air lubrication
b) Pressure measurement
c) Fuel injection
d) Sound absorption

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The function of integrator is _____.
- Q.12 Stability is defined as _____.
- Q.13 Give the name of one warning signal.
- Q.14 What do you understand by redundant?
- Q.15 Give the function of life rafts.
- Q.16 Name two pilot personal equipments.
- Q.17 What is the need of oxygen bottle in aircraft?
- Q.18 Give one disadvantage of bladder fuel tank.
- Q.19 Lubricants is used in _____ part of aircraft.
- Q.20 De-icing works on the principle of _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe various types of aircraft fuels.
- Q.22 Write short note on retractive systems.
- Q.23 Explain the construction of contamination control filters.
- Q.24 Describe the need and working of liquid to gas converters.
- Q.25 Compare the capsules and bellows.
- Q.26 Explain the anti-skid systems of air crafts.
- Q.27 Classify the types of fuels valves and its construction.
- Q.28 What re various principle of automation?